

# Fine-mapped eQTL evaluation using XQTL mega-eQTL data

## 1 Overview

This analysis applies the variant-effect benchmark in `eqtl.tex` to six cell-type mega-eQTL fine-mapping datasets supplied as Sum of Single Effects (SuSiE) posterior inclusion probabilities (PIPs) for expression quantitative trait loci (eQTLs). In this report, XQTL names this six-group source. The groups are astrocytes (Ast), excitatory neurons (Exc), inhibitory neurons (Inh), microglia (Mic), oligodendrocyte precursor cells (OPC), and oligodendrocytes (Oli). The input does not contain an endothelial group. We compare these results with the existing singlebrain fine-mapping, using its broad Ast, Ext, IN, MG, OPC, and OD panels, respectively.

## 2 Methods

For both data sources, gene identifiers are version-normalized and matched to the same GENCODE hg38 gene annotation. Variants are restricted to single-base ACGT substitutions on the gene chromosome within 512 kb of the gene transcription start site (TSS). Variants with  $\text{PIP} > 0.75$  are positives; those with  $\text{PIP} < 0.01$  are negatives. The benchmark retains every positive and samples up to 1,000 negatives per group with a fixed seed. Each model receives a 1 Mb sequence window centered at the gene TSS.

The trained comparisons reuse the head-only, LoRA, and LoRA+LoCon checkpoints and RNA track mappings in `eqtl.tex`. Native AlphaGenome mutation effects use all available RNA-seq and CAGE tracks, followed by the same biologically constrained track selection used in that report. Performance includes area under the receiver operating characteristic curve (AUROC) and average precision (AP), overall and by distance to the TSS; random AP equals the positive prevalence within each distance bin. CAGE means Cap Analysis of Gene Expression.

Fine-mapping agreement is computed on gene-variant keys consisting of gene, chromosome, one-based position, reference allele, and alternate allele. Exact high-confidence overlap uses all eligible variants and reports intersection, Jaccard similarity, and directional recall at  $\text{PIP} > 0.75$ .

## 3 Fine-mapping agreement

The full scan covered 70.9–218.7 million input rows per group. After matching to an annotated gene, requiring a single-base substitution on the gene chromosome, and applying the TSS window, 14.3–41.2 million records per group remained. Table 1 gives counts and PIP classes.

At the high-confidence threshold, exact gene-variant Jaccard similarity is 0.080–0.118 (Table 2). XQTL positive recall against singlebrain is 0.28–0.37, while singlebrain positive recall against XQTL is 0.09–0.16. Thus the sources share some high-confidence calls but the two sets are not interchangeable.

Table 1: XQTL source counts. Eligible rows satisfy the common gene, allele, chromosome, and TSS-window filters.

| Group           | Input rows  | Eligible rows | PIP > 0.75 | PIP < 0.01 |
|-----------------|-------------|---------------|------------|------------|
| Astrocyte       | 167,399,779 | 31,850,543    | 581        | 31,769,295 |
| Excitatory      | 218,708,555 | 41,150,681    | 1,784      | 40,976,915 |
| Inhibitory      | 207,868,448 | 39,394,880    | 813        | 39,281,800 |
| Microglia       | 70,865,864  | 14,332,701    | 168        | 14,305,646 |
| OPC             | 122,683,778 | 24,317,596    | 358        | 24,269,005 |
| Oligodendrocyte | 178,552,659 | 33,830,282    | 852        | 33,738,029 |

Table 2: Exact high-PIP overlap. Recall columns use the named source as denominator.

| Group           | XQTL $n$ | Singlebrain $n$ | Shared | Jaccard | XQTL recall | SB recall |
|-----------------|----------|-----------------|--------|---------|-------------|-----------|
| Astrocyte       | 581      | 1,231           | 172    | 0.105   | 0.296       | 0.140     |
| Excitatory      | 1,784    | 3,420           | 504    | 0.107   | 0.283       | 0.147     |
| Inhibitory      | 813      | 1,838           | 236    | 0.098   | 0.290       | 0.128     |
| Microglia       | 168      | 668             | 62     | 0.080   | 0.369       | 0.093     |
| OPC             | 358      | 929             | 122    | 0.105   | 0.341       | 0.131     |
| Oligodendrocyte | 852      | 1,820           | 283    | 0.118   | 0.332       | 0.155     |

## 4 Variant-effect performance

The completed benchmark scores 10,452 of the 10,556 cached records. The remaining 104 records are variants that could not be represented in the reference sequence and are excluded consistently from model and native-track comparisons. Table 3 reports aggregate performance; the native series uses one biologically constrained native RNA-seq or CAGE track selection per XQTL panel. Head-only is the strongest trained model on the aggregate TSS and gene-span metrics. Native matched tracks are close to the trained models for gene-span average precision, while distance-stratified performance is shown in Figures 4 and 4.

Table 3: Aggregate XQTL variant-effect performance across the six panels.

| Method                | TSS AUROC | TSS AUPR | Gene AUROC | Gene AUPR |
|-----------------------|-----------|----------|------------|-----------|
| Head-only             | 0.564     | 0.514    | 0.658      | 0.626     |
| LoRA                  | 0.550     | 0.498    | 0.643      | 0.608     |
| LoRA+LoCon            | 0.556     | 0.505    | 0.635      | 0.606     |
| Native matched tracks | 0.541     | 0.491    | 0.627      | 0.626     |

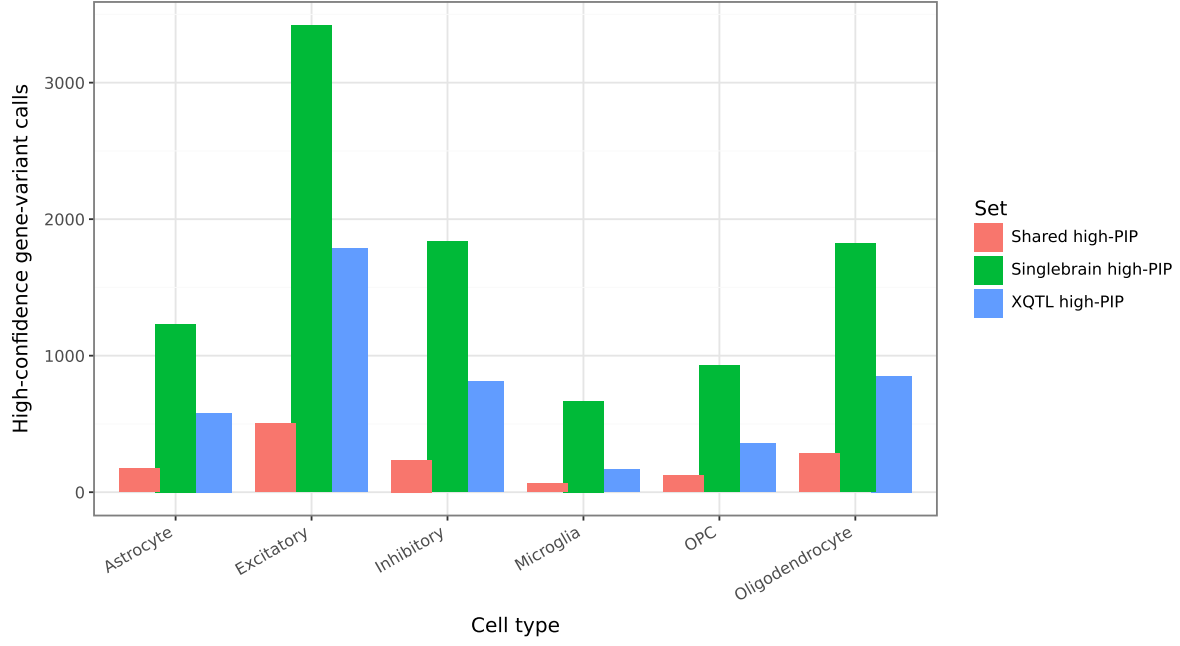


Figure 1: Counts of eligible high-PIP gene-variant calls in each source and their exact intersection.

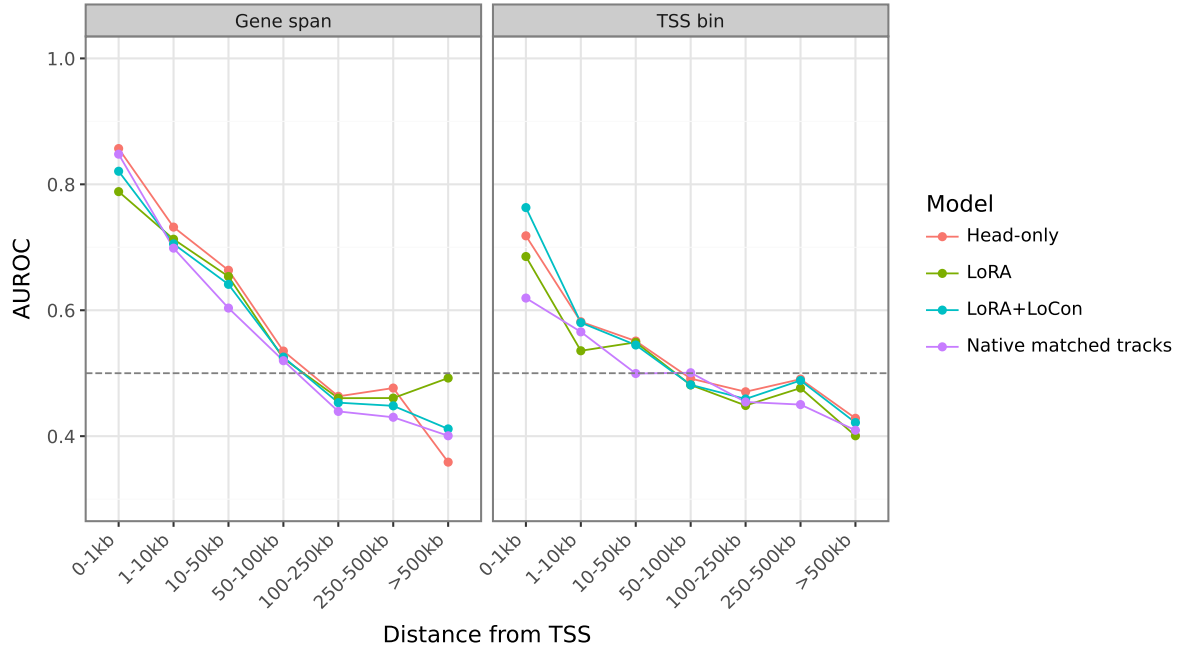


Figure 2: AUROC by absolute distance from the TSS.

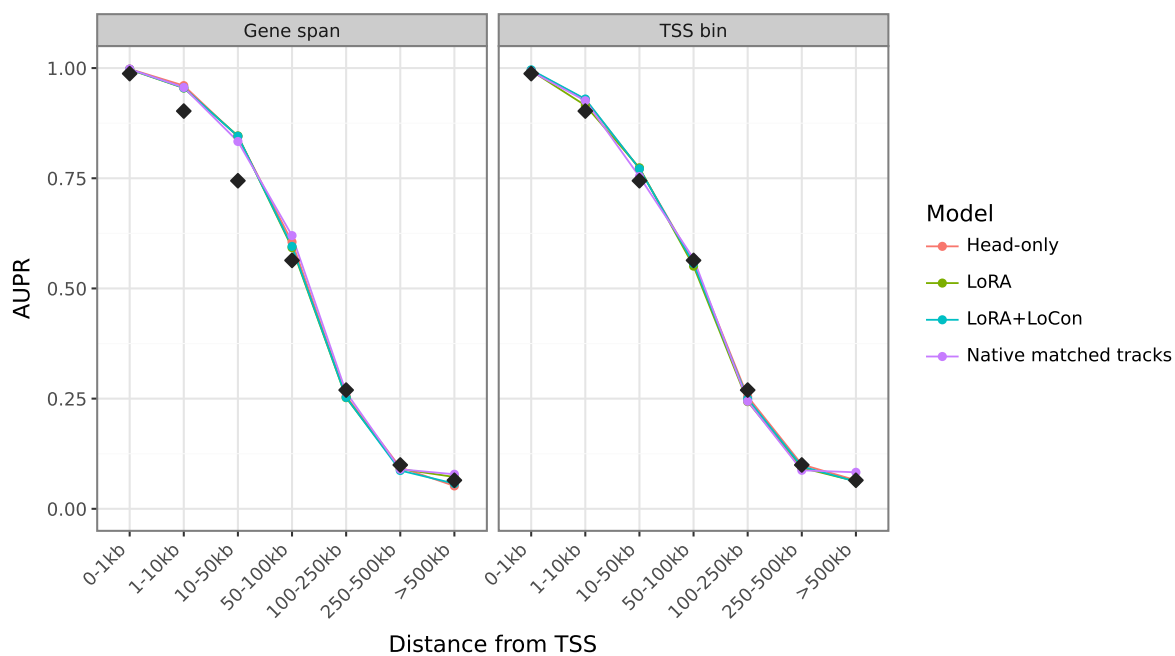


Figure 3: Average precision by absolute distance from the TSS. The random reference is the positive prevalence in each bin.